Malignant Infarction

Decompressive hemicraniectomy selection criteria, evidence, and supportive ICU care.

STAGE 1: 0 - 24 HOURS
Baseline Core & Serial NIHSS Checks

STAGE 2: 24 - 48 HOURS
Peak Edema Phase & Serial CT Scan

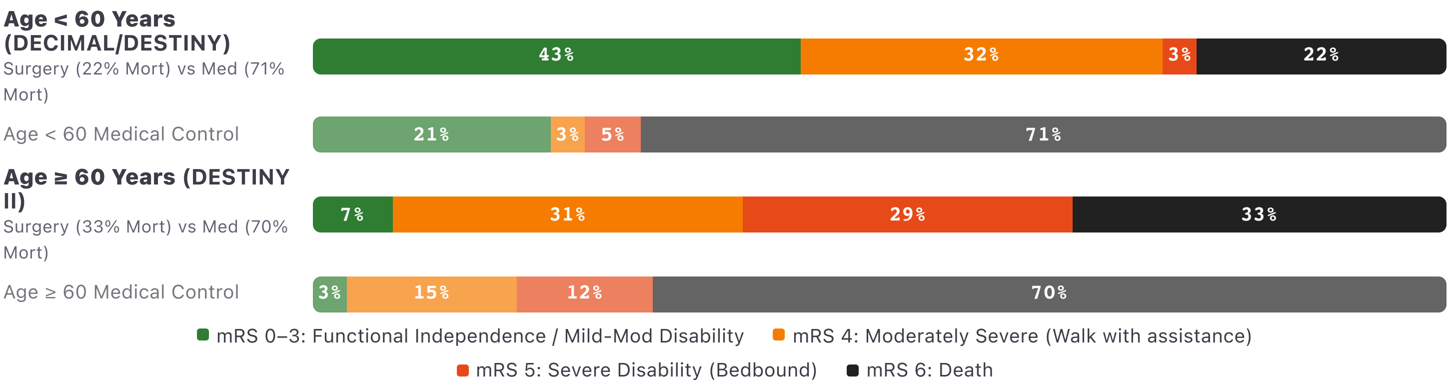
STAGE 3: < 48H SURGERY
Decompressive Hemicraniectomy

1. Decompressive Hemicraniectomy Selection Criteria

Clinical Deficit Severity:
- NIHSS **> 15** (non-dominant hemisphere)
- NIHSS **> 20** (dominant hemisphere)
- AND decrease in level of consciousness (NIHSS Item 1a score **≥ 1** / obtunded or stuporous)
- **Timing**: Surgery performed **within 48 hours** of onset.

Radiographic Markers:
- Infarction of **≥ 50%** of the MCA territory (CT/MRI)
- DWI core volume **> 82 mL** within 6 hours
- DWI core volume **> 145 mL** within 14 hours
- Midline shift or mass effect on repeat imaging
- **Surgical Spec**: Bone flap diameter **≥ 12–15 cm** with duraplasty.

2. Surgical Outcomes & Evidence (By Age Group)



• **Age < 60**: NNT = 2 for survival, NNT = 4 for mRS ≤3. | • **Age ≥ 60**: NNT = 3 for survival, NNT = 25 for mRS ≤3. *Goals-of-care discussion critical.

3. Supportive ICU Care & Medical Management

Positioning	Elevate HOB 30 degrees; maintain straight head/neck alignment to maximize venous outflow.
Fluids	Use Isotonic Saline (0.9% NS). **Avoid hypotonic fluids** (e.g. D5W, 0.45% NS, LR) which worsen edema. Maintain euvolemia.
Osmotherapy	Consider **targeted PRN** hyperosmolar agents (HTS 3% or Mannitol) for acute decline or severe mass effect. *Prophylactic osmotherapy is not recommended.*
Metabolic	Target normothermia (<37.8°C). Target normocapnia (PaCO2 35-45 mmHg); avoid hypoventilation/hypercapnia.
Steroids	**Class III (Harmful)**: Corticosteroids are NOT recommended for reducing cerebral edema in acute ischemic stroke.

DECIMAL: Vahedi K et al. *Stroke*. 2007;38:2506-2517. | **DESTINY:** Jüttler E et al. *Stroke*. 2007;38:2518-2525.
HAMLET: Hofmeijer J et al. *Lancet Neurol*. 2009;8:326-333. | **DESTINY II:** Jüttler E et al. *N Engl J Med*. 2014;370:1091-1100.
AHA Guidelines: Wijdicks EF et al. *Stroke*. 2014;45:1222-1238.